

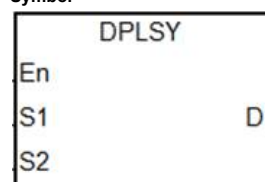
API	Instruction			Operand							Description						
2700	D	PLSY		S_1 , S_2 , D							High-speed pulse output (without ramp-up/down process)						

Device	X	Y	M	S	T	C	HC	D	FR	SM	SR	E	K	16#	"\$"	F
S_1							●	●	●		○		○	○		
S_2							●	●	●		○		○	○		
D		○														

Data type	BOOL	WORD	DWORD	LWORD	UINT	INT	DINT	LINT	REAL	LREAL	TMR	CNT	STRING
S_1			●				●						
S_2			●				●						
D	●												

Pulse Instruction	16-bit instruction	32-bit instruction
—	—	AS

Symbol



- S_1 : Pulse output frequency
- S_2 : Number of pulses to output
- D : Pulse output device

Explanation

- This instruction specifies the pulse output (S_1 , S_2) for the device specified in D with no ramp up or ramp down when the frequency changes. S_1 specifies the pulse output frequency range: for line driver output models, the range is between 0–4MHz; for open collector output models, the range is between 0 Hz–200 kHz. The unit of the output frequency is 1 Hz. The error rate for 200 kHz is about 0.02% and for 100 kHz it is about 0.01%. The error rate decreases with decreasing frequency. For example, if the output frequency setting is 199990 Hz, the actual output is 199960 Hz. If the output frequency setting is 99999 Hz, the actual output is 99990 Hz. If the output frequency is out of the valid range, the PLC automatically processes it as the maximum or minimum pulse output frequency.
- You can change the output frequency specified by S_1 during the execution of the instruction without using the ramp up/down process. The time to change the frequency is when the PLC is scanning the instruction and the instruction has completed output of a full pulse.
- S_2 specified the number of output pulses. The range is between 0–2,147,483,647. When you set the number of output pulses to 0, then the number of pulses is not restricted and pulses are output constantly until the instruction is disabled. When the number of output pulses is set to less than 0, no pulses are output.
- After the instruction starts to output pulses, the number of output pulses specified by S_2 cannot be changed.
- D only allows Y0.0–Y0.11 as the output devices. After the instruction is executed, its output function becomes the high-speed output, and the basic instruction output point control is invalid. It is suggested that you do not use the general output function after using the high-speed output function in the program.
- The ratio of Duty-OFF Time and Duty-ON Time for the pulse output is 1:1.
- There is no limit to how many times you can use the instruction a program. But only one output instruction that uses the same output point can be executed in the same scan cycle. When several high-speed pulse output instructions start the same output point in the program, the PLC first performs the output based on the instruction that is executed first.
- After every high-speed output instruction for every output point is executed, other instruction occupying the same output point can not start until the high-speed output instruction in execution is disabled.
- When the high-speed output instruction is executed in an interrupt program or is not in the main process, it is suggested that you use the instruction with the auto-reset function when output is completed and the PLC updates the output state in the END instruction.
- After the stop flag is set, the PLC stops the output and clears the busy flag only after executing the start instruction a second time and outputting a full pulse. The PLC continues to output pulses when the stop flag is reset and the previously stopped pulse output is not finished. You set and reset the stop flag is set in the program.
- The high-speed output points and corresponding SM/SR are listed in the following table.

Output point number	Attribute#2	Y0.0	Y0.1	Y0.2	Y0.3	Y0.4	Y0.5
Busy flag	R	SM460	SM472	SM480	SM492	SM500	SM512
Completion flag#3	R/W	SM461	SM473	SM481	SM493	SM501	SM513
Stop flag	R/W	SM463	SM474	SM483	SM494	SM503	SM514
Output completion auto-reset#4	R/W	SM470	SM475	SM490	SM495	SM510	SM515
Present output position#1 (32-bit)	R/W	SR460 SR461	SR474 SR475	SR480 SR481	SR494 SR495	SR500 SR501	SR514 SR515

Output point number	Attribute #2	Y0.6	Y0.7	Y0.8	Y0.9	Y0.10	Y0.11
Busy flag	R	SM520	SM532	SM540	SM552	SM560	SM572
Completion flag #3	R/W	SM521	SM533	SM541	SM553	SM561	SM573
Stop flag	R/W	SM523	SM534	SM543	SM554	SM563	SM574
Output completion auto-reset #4	R/W	SM530	SM535	SM550	SM555	SM570	SM575
Present output position #1 (32-bit)	R/W	SR520 SR521	SR534 SR535	SR540 SR541	SR554 SR555	SR560 SR561	SR574 SR575

Notes:

#1: All present output positions are latched when power is off.

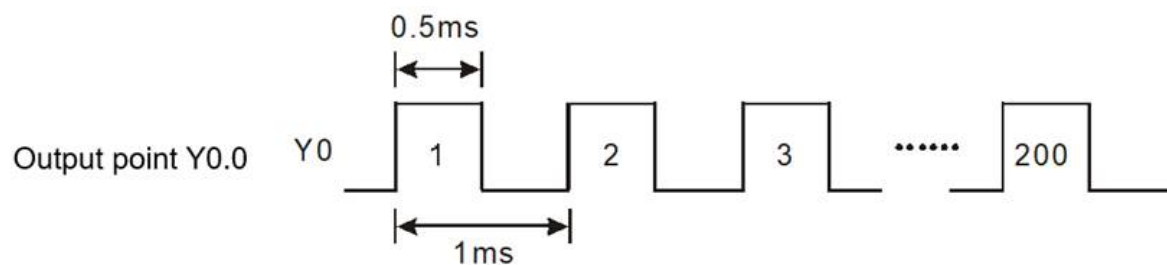
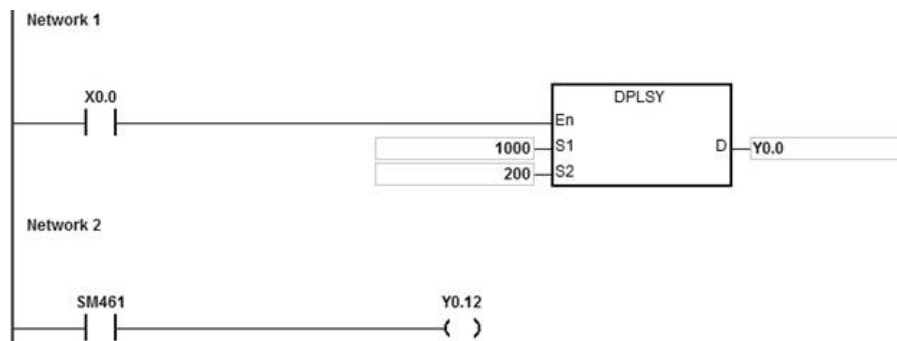
#2: R means Read-only registers, and the data in the registers cannot be modified. R/W means the data in the registers can be read and written.

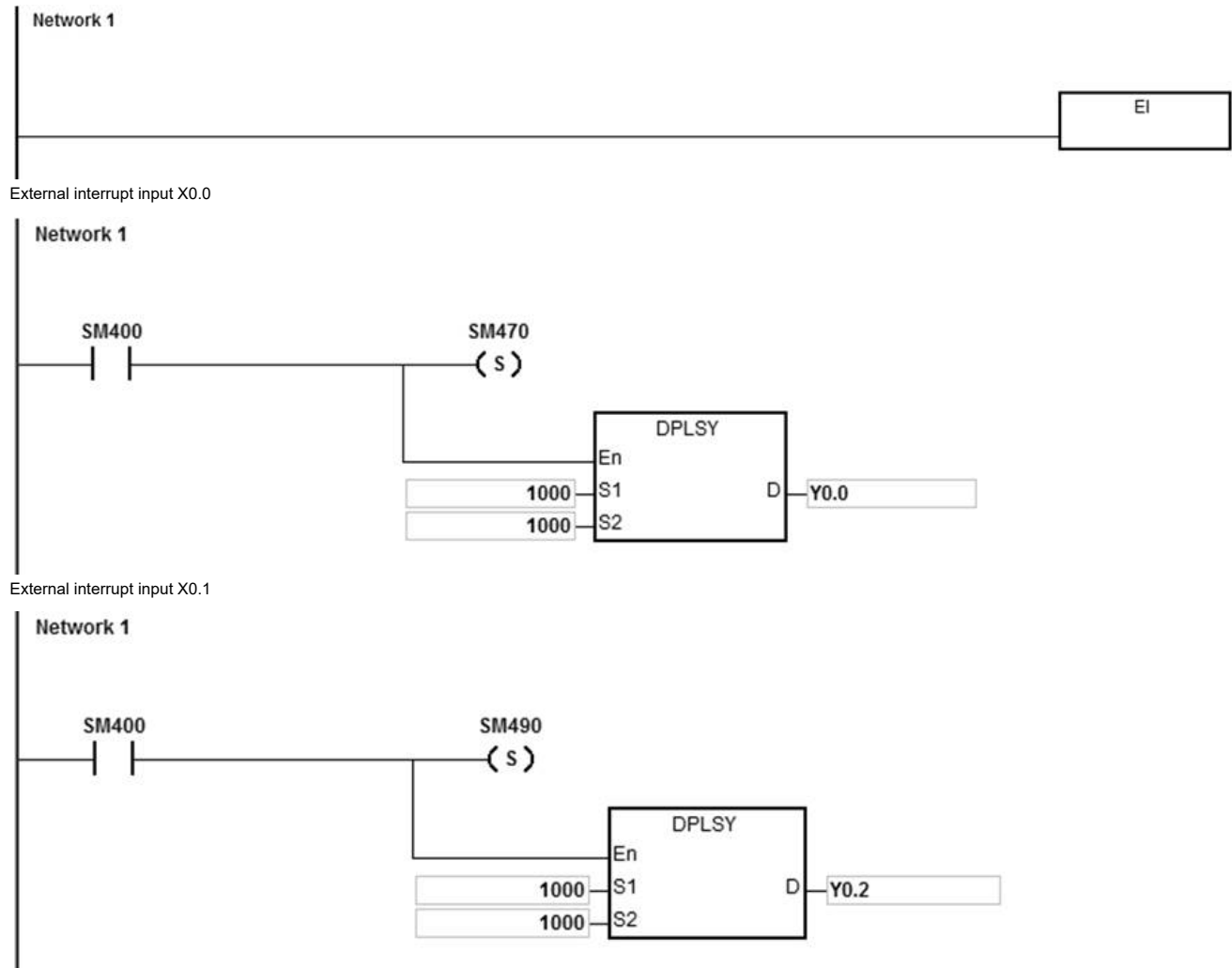
#3: It is suggested that you clear the completion flag. If the completion flag is not cleared, it is cleared automatically when the high-speed output instruction is executed the next time.

#4: You set the output completion auto-reset flag. The PLC clears the flag automatically after the output is completed.

Example 1

- When X0.0 is ON, Y0.0 outputs 200 pulses at 1kHz. SM461 is ON when the pulse output is completed and then Y0.12 is ON.
- When X0.0 is OFF, Y0.0 stops the output. It restarts the pulse output when X0.0 switches to ON again.



Example 2**Explanation**

1. Y0.0 outputs 1000 pulses when X0.0 receives one external interrupt signal. Y0.2 outputs 1000 pulses when X0.1 receives one external interrupt signal.
2. When the external interrupt input X triggers the pulse output from Y, the interval time between the Y pulse output completion and the next external interrupt input X trigger must be one or more PLC scan cycles.

Example 3 (ST program)

```

0001 IF M0 THEN
0002     DPLSY (1000,1000,Y0.0);
0003     M1 := TRUE;
0004 ELSIF M1 AND SM461 THEN
0005     SM470 := TRUE;
0006     M1 := FALSE;
0007 END_IF;

```

Explanation

1. When M0 is ON, Y0.0 outputs 1000 pulses at 1kHz. When M1 is ON, it indicates Y0.0 is used for high-speed pulse output.
2. When the pulse output is completed, SM461 is ON. And if M1 is also ON, it triggers SM470 (output completion auto-reset for Y0.0) to be ON and Y0.0 is free and ready to be used again.
3. When M0 switches from OFF to ON, the pulse output is restarted.
4. When there are more than two high-speed output instructions in a program, do not use the variable name, M1, repeatedly.
5. It is suggested not to use ST language in a program if there is any chance the outputting needed to be stopped.